REV A LAYOUT REVIEW. NOT FOR FABRICATION.

915 MHz LoRa card

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

J1
TSW-110-07-G-S
GND 1
SLOT_5V 2
SLOT_5V 3
GND 4
SPI_SCK 5
6 SPI_MOSI
7 SPI_MISO
8 CS_N
9 RESET_N
10 GND

J2
TSW-108-07-G-S
BUSY 1
IRQ 2
TX_EN 3
RX_EN 4
5 AUX0
6
7 IOREF_3V3
8 GND

U1
E22-900M30S
GND 1
GND 2
GND 3
GND 4
GND 5
RX_EN 6
TX_EN 7
AUX0 8
SLOT_5V 9
SLOT_5V 10
GND 11
12 GND
13 IRQ
14 BUSY
15 RESET_N
16 SPI_MISO_SRC
17 SPI_MOSI
18 SPI_SCK
19 CS_N
20 GND
21
22 GND

Module pad 21 ANT: NC, use module IPEX. Both pads 9 and 10 are VCC. H1 is the mandatory alignment-post hole.
DIO2 is AUX0 for a local test point only: do not connect it to a Pico GPIO.
AUX1 is reserved. DIO3 is internal TCXO control, configured to 2.2 V.

R5
22
SPI_MISO_SRC 1
2 SPI_MISO

TP1
SPI_SCK
SPI_SCK 1
PROBE

TP2
SPI_MOSI
SPI_MOSI 1
PROBE

TP3
SPI_MISO
SPI_MISO 1
PROBE

TP4
CS_N
CS_N 1
PROBE

#FLG01
POWER DECLARATION: SLOT_5V
SLOT_5V 1
POWER

#FLG02
POWER DECLARATION: GND
GND 1
POWER

#FLG03
POWER DECLARATION: IOREF_3V3
IOREF_3V3 1
POWER

TP5
RESET_N
RESET_N 1
PROBE

TP6
BUSY
BUSY 1
PROBE

C1
220uF
SLOT_5V 1
+
-
2 GND

C2
22uF
SLOT_5V 1
1
2
2 GND

C3
100nF
SLOT_5V 1
1
2
2 GND

TP7
IRQ
IRQ 1
PROBE

TP8
TX_EN
TX_EN 1
PROBE

TP9
RX_EN
RX_EN 1
PROBE

TP10
IN_5V
SLOT_5V 1
PROBE

R1
10k
IOREF_3V3 1
1
2
2 CS_N

R2
10k
IOREF_3V3 1
1
2
2 RESET_N

TP11
CLIP_GND
GND 1
PROBE

TP12
AUX0
AUX0 1
PROBE

R3
10k
TX_EN 1
1
2
2 GND

R4
10k
RX_EN 1
1
2
2 GND

H1
M3 alignment hole

TP13
MOD_5V
SLOT_5V 1
PROBE

TP14
MOD_GND
GND 1
PROBE

TP15
IOREF
IOREF_3V3 1
PROBE

MISO: source -> 22R -> slot A7. Fit OR option: scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

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Title: 915 MHz LoRa card

Size: A2	Date: 2026-09-17	**Rev: A-schematic**
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